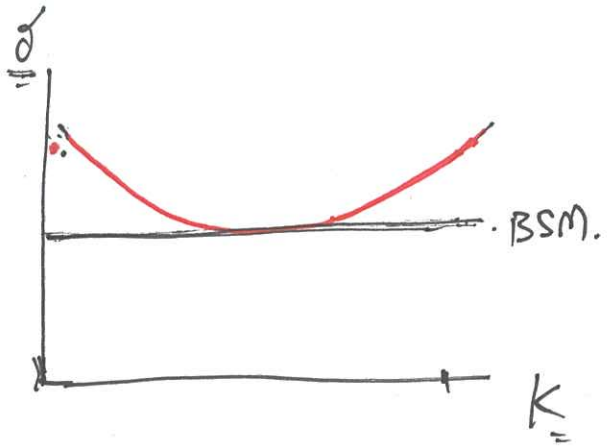
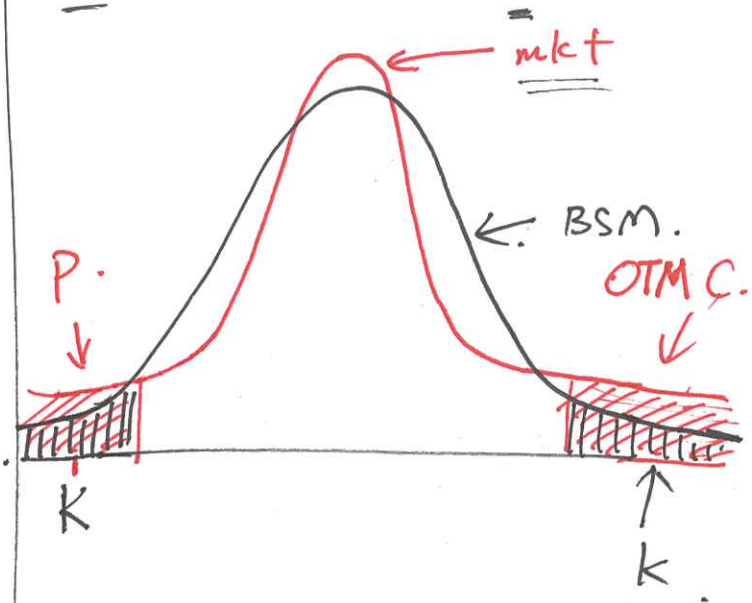
$C = (S_0, K, T, r, \sigma)$.

underlying asset.

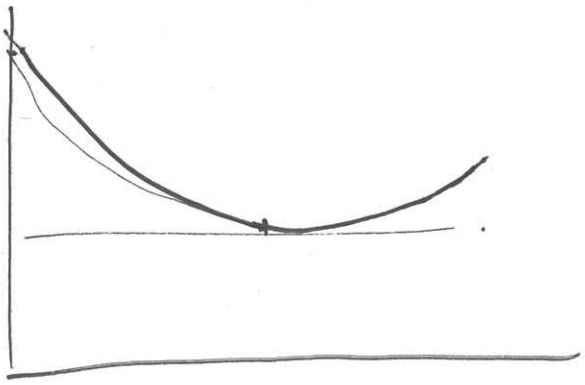
BHP.

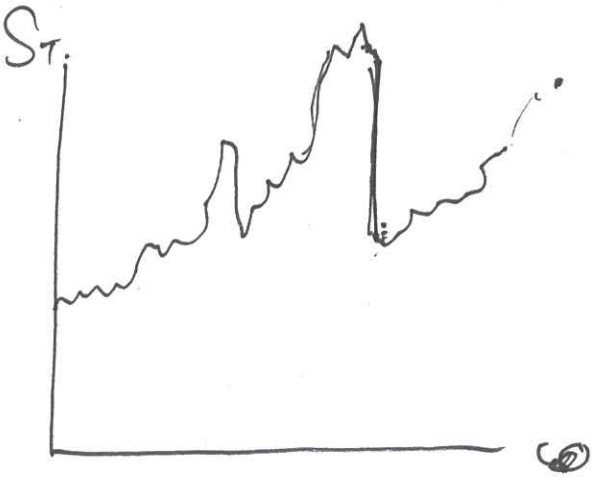


BSM: constant σ.



Equity.





stock A : pay div @ q.

stock B : no div.

$r_A = r_B = g_B$

$\| \quad \| \quad \|$

$g_A + q_A$

$P = \frac{D}{r-g}$

$r - g = \frac{D}{P} = q.$

$r = g + q$

A : $S_0 \longrightarrow S_{T_A}$

B : $S_0 \longrightarrow S_{T_A} \cdot e^{q_A \cdot T}$

$S_{TA} = S_0 \cdot e^{g_A \cdot T}$

$S_{TB} = S_0 \cdot e^{g_B \cdot T} = S_0 \cdot e^{(g_A + q_A) \cdot T} = S_0 \cdot e^{g_A \cdot T} \cdot e^{q_A \cdot T}$

$= S_{TA} \cdot e^{q_A \cdot T}$

$S_0 \cdot e^{-qT} \longrightarrow S_T$

Replace S_0 with $S_0 \cdot e^{-qT}$.

	t_0	T	
		$S_T > K$	$S_T \le K$
A	C $K \cdot e^{-rT}$ Cash / Bond.	$S_T - K$ K	0 K
	$C + K \cdot e^{-rT}$	S_T	K
B.	$e^{-qT} \cdot S_0$, with ↑ # of shares div. reinvestment	S_T	S_T

no reinvestment. $S_0 \rightarrow S_T$

$$e^{-qT} \cdot S_0 \rightarrow e^{-qT} \cdot S_T$$

with reinvestment. $e^{-qT} \cdot S_0 \rightarrow e^{-qT} \cdot S_T \cdot \boxed{e^{qT}}$

At T, $A \ge B$.

At t_0, $C + K \cdot e^{-rT} \ge S_0 \cdot e^{-qT}$

Put - call Parity

$$c + k \cdot e^{-rT} = P + S_0 \cdot e^{-qT}$$

	t_0	T	
A	c	$S_T > K$	$S_T \leq K$
	$K \cdot e^{-rT}$	$S_T - K$	0
		K	K
Net	$c + K \cdot e^{-rT}$	S_T	K
C	P	0	$K - S_T$
	$e^{-qT} \cdot S_0$	S_T	S_T
	$P + e^{-qT} \cdot S_0$	S_T	K

At T, $\begin{cases} S_T > K, & A = C \\ S_T \leq K, & A = C \end{cases}$

$\Rightarrow c + K \cdot e^{-rT} = P + e^{-qT} \cdot S$

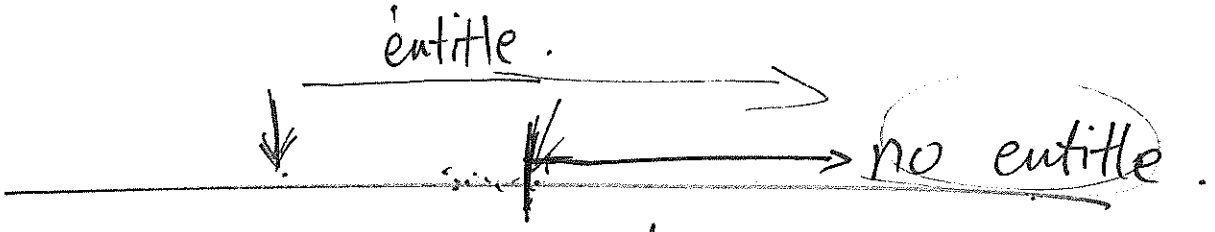
$$P \cdot S_u + (1-P) S_d = S_0 \cdot e^{r\Delta t}.$$

$S_0 \cdot u \qquad S_0 \cdot d.$

$$P = \frac{e^{r\Delta t} - d}{u - d}.$$

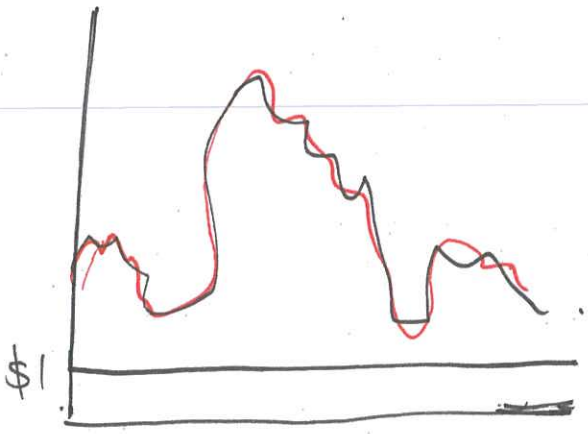
$r = q$

$r - q.$



ex- div:

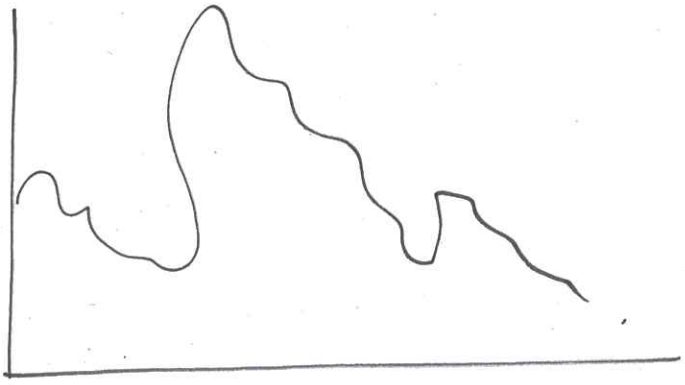
↓ D.



riskless = D = $1.

risky ↑ by $2.

$S_0 = \$20$ S↑ by $2.



S_0.

Replace S_0 with $S_0 \cdot e^{-qT}$

Replace S_0 with $S_0 - PV(D)$.

